

SehatMitra-AI

Smart Rural Healthcare Assistant

Multilingual voice triage, edge-resilient AI diagnostics, and ABDM-compliant digital health records — engineered for India's last-mile Primary Health Centres.

 Voice-First UX

 Groq LPU Triage

 Vision Diagnostics

 ABHA Identity

 Edge-DB Resilience

CORE MISSION Deliver doctor-grade triage and health record continuity to every village PHC — regardless of language, literacy, or network strength.

PRESENTED BY
Team SehatMitra-AI
Principal Architect & Engineering Team

Four Critical Rural Healthcare Failures

01



Illiteracy & Linguistic Diversity

Patients cannot read or type in English or formal Hindi — text-first health apps exclude the majority of rural users.

02



Shortage of Primary Doctors

Remote PHCs run 1 doctor per thousands of patients. Delayed triage routinely escalates into avoidable emergencies.

03



Incomprehensible Lab Reports

Blood, lipid, and urine reports arrive in dense medical jargon — causing panic instead of informed action.

04

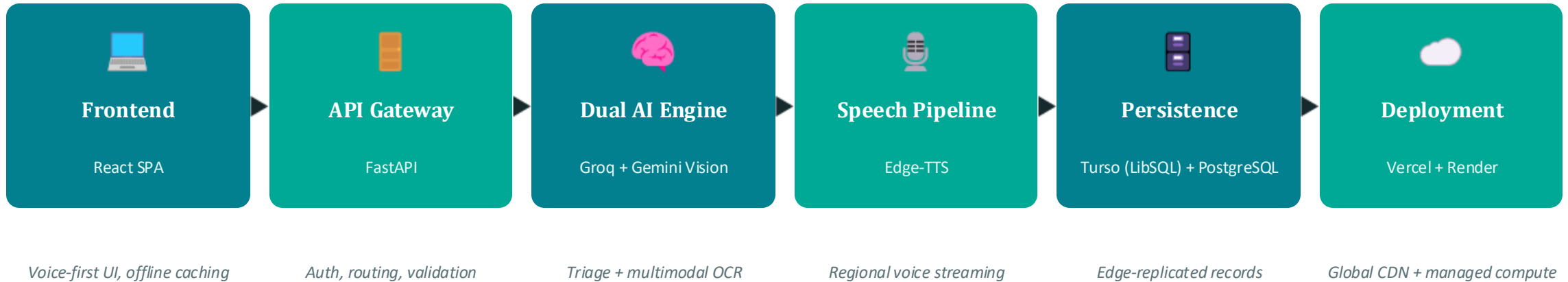


Missing Medical Records

Paper prescriptions are lost between visits — no persistent, longitudinal health history follows the patient.

Net effect: patients disengage, disease presents late, and there is no digital trail to learn from.

High-Level System & Data Flow Architecture



⚙️ DESIGNED FOR 2G / 3G RURAL NETWORKS

Turso (LibSQL) edge replicas keep read/write latency low even on high-latency rural links, syncing back to primary on reconnect.

Groq LPU inference returns sub-second structured responses, minimizing payload round-trips over constrained bandwidth.

Edge-TTS streams audio incrementally so consultations remain audible even under packet loss.

Illiteracy & Regional Dialects

THE RURAL BOTTLENECK

PROBLEM 1 · LANGUAGE ACCESS Illiteracy & Regional Dialects THE RURAL BOTTLENECK Browser-native SpeechSynthesis fails for low-resource Indian languages — voices are robotic, mispronounced, or missing, completely locking out illiterate patients.

THE ENGINEERING SOLUTION

Decoupled backend audio streaming pipeline that generates and streams neural Indic voices (edge-tts) directly to the device with sub-600ms latency.

SPECIFIC TECH STACK USED

- edge-tts → pa-IN-OjasviNeural (Punjabi)
- edge-tts → or-IN-SubhasiniNeural (Odia)
- edge-tts → hi-IN-MadhurNeural (Hindi)

CLINICAL / USER IMPACT

100% hands-free voice consultation — patients speak and listen in their native dialect from intake to triage report.



Voice-Based Clinical Triage

Speak or type your symptoms. SehatMitra-AI will instantly evaluate your clinical risk level.

SELECT PREFERRED LANGUAGE

English हिन्दी बांग्ला తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ മലയാളം
ਪੰਜਾਬੀ ଓଡ଼ିଆ ગુજો

Current Locale: en-IN

Idle

INITIAL_INTAKE_PROMPT

Hello! I am SehatMitra, your clinical voice assistant. How are you feeling today? Please describe your symptoms.

🔊 🗣️ replay_question



Tap to speak symptoms

Your Answer

Your transcribed speech or typed answer will appear here...

Submit Answer



Primary Doctor Shortage & Slow Diagnosis

THE RURAL BOTTLENECK

Standard LLMs hallucinate medical advice and carry high inference latency — unacceptable when running over degraded 2G/3G rural networks with a patient waiting in real time.

THE ENGINEERING SOLUTION

Groq LPU inference executes a structured SOCRATES clinical triage protocol (Site, Onset, Character, Radiation, Associations, Time, Exacerbating factors, Severity) with automatic model failover for resilience.

SPECIFIC TECH STACK USED

groq SDK → primary: Llama 3.3
fallback: Qwen 2.5 (dynamic failover)

CLINICAL / USER IMPACT

Sub-second response times with a structured, explainable Risk Badge — Mild / Moderate / Severe — at every consultation.

 Sub-Second Risk Badge (Groq LPU)



SM

Symptom Chat

AI Powered Triage Assistant

Reset

Do you have any nausea, vomiting, or abdominal pain?

 Speak

12:17 pm

no

12:17 pm

Based on your symptoms of high fever, severe headache, body chills, and joint pain, it is important to rule out conditions like dengue, malaria, or a severe viral infection. Please see a General Physician for a physical examination and blood tests (such as CBC and Dengue/Malaria tests) as soon as possible today. If you develop a stiff neck, confusion, severe abdominal pain, or bleeding, go to the emergency room immediately.

 Speak

12:17 pm

Chat Interface

 Describe your symptoms in detail...
 

HIGH RISK

Primary Health Risk Report

Generated: Just now

REASONS

- High-grade fever (102°F) persisting for 3 days.
- Severe headache rated 6/10.
- Associated symptoms include body chills and joint pain.
- Absence of gastrointestinal symptoms (nausea/vomiting) or abdominal pain.
- Symptom cluster raises suspicion for viral infections such as Dengue, Malaria, or Influenza.

RECOMMENDATION

Seek medical attention today for a General Physician consultation. Blood tests (CBC, Platelet count, Malaria/Dengue antigens) are likely required. Drink plenty of fluids to stay hydrated. Use paracetamol for fever and headache as per dosage instructions. Monitor for red flags: stiff neck, confusion, severe abdominal pain, bleeding gums, or rash. If any red flags appear, seek emergency care immediately.

CHECKLIST FOR DOCTOR / PATIENT

- ☐ Check for meningeal signs (stiff neck, Kernig's sign) to rule out meningitis.
- ☐ Perform complete blood count (CBC) to check platelet levels and white blood cell count.
- ☐ Test for Malaria (Peripheral smear/RDT) and Dengue (NS1/IgM/IgG) based on local prevalence.
- ☐ Assess hydration status and vital signs (BP, Pulse, SpO2).
- ☐ Evaluate for any rash or petechiae.

 This AI triage is for informational purposes only and does not replace professional medical advice, diagnosis, or treatment. Always seek the advice of your physician or other qualified health provider with any questions you may have regarding a medical condition.

Download Doctor Slip (PDF)

SehatMitra-AI | Smart Rural Healthcare Assistant

SOCRATES Clinical Checklist
(ZeroHallucination)

05 / 10

Medical Jargon in Lab Reports

THE RURAL BOTTLENECK

Rural patients cannot interpret blood tests, lipid profiles, or urine reports — dense clinical terminology and unfamiliar units turn a routine result into a source of fear or dangerous inaction.

THE ENGINEERING SOLUTION

A multimodal Vision OCR pipeline extracts biomarkers directly from a photographed report, normalizes noisy scans, and generates a plain-language explanation with out-of-range values flagged.

SPECIFIC TECH STACK USED

google-genai (Gemini Vision) → multimodal OCR
Pillow + NumPy → image pre-processing

CLINICAL / USER IMPACT

Plain-language biomarker explanations in 12 Indian languages, with abnormal values automatically flagged for follow-up.



Multimodal Medical Report Analyzer

Capture or upload your blood tests, prescriptions, or report images. SehatMitra-AI will parse key medical biomarkers.

Biomarker Summary

[Listen Summary \(सुनी\)](#)

Your blood test results are mostly normal, which is good news. Your red blood cell count and hematocrit are just a little above the usual range, and the MCHC is slightly low, but these small differences are often not a cause for concern and can be managed with simple lifestyle adjustments. Overall, there is no indication of any serious problem.

Extracted Health Biomarkers

WBC**NORMAL****6.1** $10^3/\mu\text{L}$

Normal Range: 4.0 - 11.0

Lym %**NORMAL****24.1** %

Normal Range: 16.0 - 43.3

HCT**HIGH****46.5** %

Normal Range: 36.0 - 46.0

RBC**HIGH****5.21** $10^6/\mu\text{L}$

Normal Range: 3.90 - 5.20

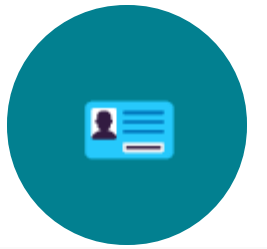


Assistance for Doctor Consultation

Please note that this is an AI-generated summary. Always share these reports and consult with your doctor.

[Download Consultation Slip](#)[Upload Another Report](#)

Lost Physical Records & Identity



THE RURAL BOTTLENECK

Physical prescriptions are easily lost between visits, and rural patients have no linkage into India's national digital health registries — every visit starts from zero history.

THE ENGINEERING SOLUTION

An ABDM-compliant digital ABHA health card is generated per patient, backed by an edge-replicated database that keeps consultation and lab history durable and instantly retrievable.

SPECIFIC TECH STACK USED

Turso (LibSQL Edge DB) → low-latency edge reads
SQLAlchemy ORM + PostgreSQL → pool_pre_ping=True,
rollback resilience

CLINICAL / USER IMPACT

Permanent consultation transcripts and lab history, with instant QR-based ABHA verification at any PHC.

My Medical Profile

Manage your ABHA digital health record card, secure your password, and verify patient metadata.



Secured Medical Records Card

[Edit Profile](#)

FULL NAME	EMAIL ADDRESS
sehatmita.ai	sehatmita.ai@gmail.com
AGE	GENDER
N/A Yrs	N/A
BLOOD GROUP	PHONE NUMBER
N/A	N/A

2G/3G Network Drops & Token Invalidation



THE RURAL BOTTLENECK

Rural connection drops trigger 401 token invalidations mid-session, cascading into forced logouts and lost triage progress right when the patient needs continuity most.

THE ENGINEERING SOLUTION

A centralized Axios request interceptor pairs with fault-tolerant JWT extraction and Turso's edge replication to serve cached state locally and resync silently the moment connectivity returns.

SPECIFIC TECH STACK USED

Firebase Auth + PyJWT → token lifecycle
 Custom FastAPI Security deps.py → resilient auth
 Turso replication → offline-cached fallback reads

CLINICAL / USER IMPACT

Zero forced logouts on flaky rural networks — sessions resume exactly where they dropped, with no data loss.

Your Health History

Your past consultations and uploaded lab report history details are secure here.

[📶 Consultation History \(3\)](#)
[📄 Report History \(2\)](#)

📅 27 August 2026 ⌚ 06:47 am

HIGH RISK

CHIEF COMPLAINT: I have had a high-grade fever of 102°F

DOCTOR RECOMMENDATION

Seek medical attention today for a General Physician consultation. Blood tests (CBC, Platelet count, Malaria/Dengue antigens) are likely required. Drink plenty of fluids to stay hydrated. Use paracetamol for fever and headache as per dosage instructions. Monitor for red flags: stiff neck, confusion, severe abdominal pain, bleeding gums, or rash. If any red flags appear, seek emergency care immediately.

SYMPTOMS ANALYSIS

- High-grade fever (102°F) persisting for 3 days.
- Severe headache rated 6/10.
- Associated symptoms include body chills and joint pain.
- Absence of gastrointestinal symptoms (nausea/vomiting) or abdominal pain.
- Symptom cluster raises suspicion for viral infections such as Dengue, Malaria, or Influenza.

💬 Open Chat

📄 Download Slip

ASHA Worker Portal & Village Epidemic Analytics



Offline-First Triage Tracking

ASHA workers log symptoms and triage outcomes in the field with zero connectivity; syncs automatically once online.



Village-Level Epidemiological Heatmaps

Aggregated, anonymized case data renders live outbreak heatmaps at the village and block level.



Emergency PHC Dispatch Alerts

Severe risk-badge cases auto-notify the nearest PHC with patient context for faster escalation.

Empowers grassroots healthcare workers with AI decision support — extending doctor-grade triage to the front line of rural care.

ASHA Worker & Community Health Surveillance Portal

Real-time disease surveillance, syndromic outbreak alerts, and clinical risk metrics across monitored villages.

[Export Epidemiological Data \(CSV\)](#)



Syndromic Outbreak Alert: Fever & Joint Pain Cluster (Dengue/Malaria Warning)
Detected in Rampur Sector 4 • Immediate Intervention Required
Recommendation: Initiate anti-larval spray & rapid Dengue NS1 testing.

WARNING



Syndromic Outbreak Alert: Diarrhea Cluster (Gastroenteritis Outbreak Risk)
Detected in Gopalpur • Immediate Intervention Required
Recommendation: Distribute ORS/Zinc packets and chlorinate local wells.

HIGH ALERT



TOTAL SCREENINGS
3



CRITICAL REFERRALS
5



MONITORED VILLAGES
8



FOLLOW-UP COMPLETE
100%

Tool-to-Problem Mapping & Demo Summary

Technical Component	Problem Solved	Clinical / User Impact
Edge-TTS (regional neural voices)	Illiteracy & dialect exclusion	Hands-free voice consultation in mother tongue
Groq LPU + SOCRATES protocol	Doctor shortage, slow diagnosis	Sub-second, structured, explainable triage
Gemini Vision + Pillow/NumPy OCR	Jargon-heavy lab reports	Plain-language biomarker alerts, 12 languages
Turso (LibSQL Edge DB) + ABHA card	Lost records, no identity	Durable longitudinal history, instant QR verify
Axios Interceptor + PyJWT + FastAPI deps	Network drops, 401 cascades	Zero forced logouts on 2G/3G networks

NEXT STEPS — FUTURE ROADMAP



Offline Edge AI Device

Raspberry Pi + ONNX-optimized model deployment for zero-connectivity PHCs.



WhatsApp Bot Integration

Bring voice triage and ABHA lookup into the channel patients already use daily.